

El Niño and the Ski Season

What the Oceanic Niño Index actually predicts about powder days, storm frequency and base depth across 42 ski regions of the western United States — and, just as usefully, where it predicts nothing at all.

932 SNOTEL STATIONS	40.2M DAILY OBSERVATIONS	1964–2026 WATER YEARS
42 SKI REGIONS	971 TESTS RUN	51 SURVIVE FDR

Prepared 21 September 2026 · Data: NOAA CPC · NRCS AWDB · NCEI nClimDiv · Method after Brown & Harper (2026), HESS 30, 5735

THE SHORT VERSION

Four findings

1. The effect is real but geographically narrow. Of 971 tests across every region, window of winter and measure, 51 survive false-discovery control. Nearly all of them sit in the northern Rockies and the Cascades, where El Niño costs roughly a foot of snow, or in New Mexico and Arizona, where it adds snow in spring.

2. For most of the West, El Niño predicts nothing. The Wasatch, the I-70 corridor, Tahoe, the San Juans, Sun Valley and the Elk Mountains never clear significance on any measure in any window. That is a finding, not a gap.

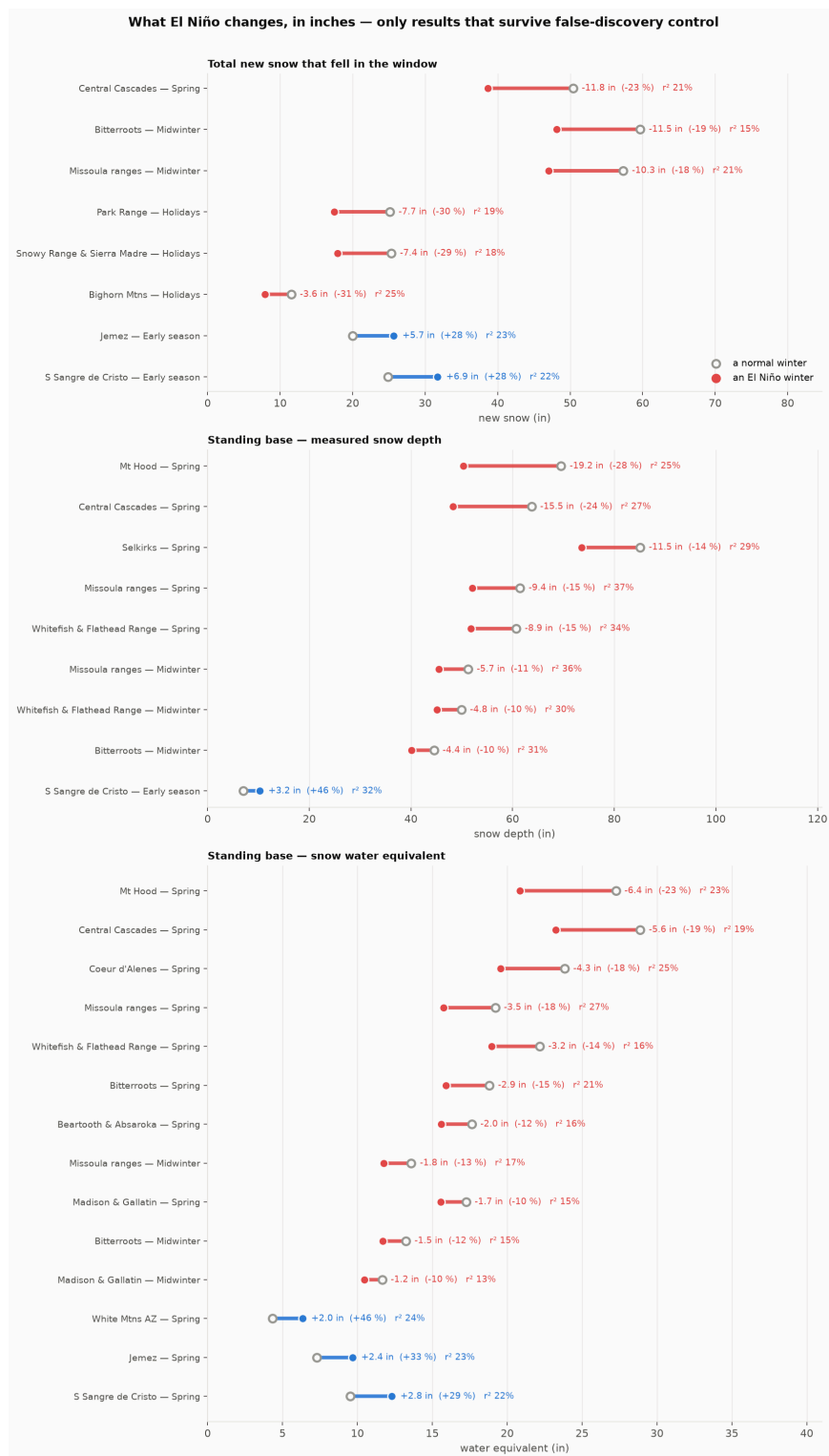
3. The Christmas–New Year window behaves differently from the rest of winter. Every other window follows a north–south seesaw. The holidays lean toward less snow in 103 of 123 tests, in all three latitude bands at once, including the southern ranges that otherwise gain.

4. The strength of an event carries no information. Tested directly across 2619 region-window-measure combinations, nothing survives. A very strong El Niño delivers more powder days than a strong one.

What survives, in inches

Inches. New snow is what fell during the window; base is what was standing, measured both as depth and as water equivalent.								
Ski Region	Window	Measure	Winters	Normal	El Niño	Difference	Change	R²
Bighorn Mtns (Meadowlark)	Holidays	Total new snow	48	11.5	7.9	-3.6 in	-31%	25%
Park Range (Steamboat)	Holidays	Total new snow	47	25.1	17.5	-7.7 in	-30%	19%
Snowy Range & Sierra Madre	Holidays	Total new snow	48	25.3	17.9	-7.4 in	-29%	18%
Mt Hood	Spring	Base, snow depth	29	69.5	50.4	-19.2 in	-28%	25%
Central Cascades (Stevens/Snoqualmie)	Spring	Base, snow depth	28	63.7	48.2	-15.5 in	-24%	27%
Mt Hood	Spring	Base, water equivalent	48	27.3	20.8	-6.4 in	-23%	23%
Central Cascades (Stevens/Snoqualmie)	Spring	Total new snow	45	50.4	38.6	-11.8 in	-23%	21%
Central Cascades (Stevens/Snoqualmie)	Spring	Base, water equivalent	45	28.9	23.3	-5.6 in	-19%	19%
Bitterroots (Lost Trail)	Midwinter	Total new snow	59	59.7	48.1	-11.5 in	-19%	15%
Missoula ranges (Montana Snowbowl)	Spring	Base, water equivalent	46	19.2	15.7	-3.5 in	-18%	27%
Coeur d'Alenes (Silver/Lookout)	Spring	Base, water equivalent	44	23.8	19.6	-4.3 in	-18%	25%
Missoula ranges (Montana Snowbowl)	Midwinter	Total new snow	46	57.3	47.1	-10.3 in	-18%	21%
Bitterroots (Lost Trail)	Spring	Base, water equivalent	59	18.8	15.9	-2.9 in	-15%	21%
Missoula ranges (Montana Snowbowl)	Spring	Base, snow depth	25	61.4	52.0	-9.4 in	-15%	37%
Whitefish & Flathead Range	Spring	Base, snow depth	25	60.7	51.8	-8.9 in	-15%	34%
Whitefish & Flathead Range	Spring	Base, water equivalent	52	22.2	19.0	-3.2 in	-14%	16%
Selkirks (Schweitzer)	Spring	Base, snow depth	25	85.1	73.6	-11.5 in	-14%	29%
Missoula ranges (Montana Snowbowl)	Midwinter	Base, water equivalent	46	13.6	11.8	-1.8 in	-13%	17%
Bitterroots (Lost Trail)	Midwinter	Base, water equivalent	59	13.2	11.7	-1.5 in	-12%	15%
Beartooth & Absaroka (Red Lodge)	Spring	Base, water equivalent	53	17.6	15.6	-2.0 in	-12%	16%
Missoula ranges (Montana Snowbowl)	Midwinter	Base, snow depth	25	51.2	45.5	-5.7 in	-11%	36%
Madison & Gallatin (Big Sky)	Midwinter	Base, water equivalent	60	11.6	10.5	-1.2 in	-10%	13%
Bitterroots (Lost Trail)	Midwinter	Base, snow depth	25	44.5	40.1	-4.4 in	-10%	31%
Madison & Gallatin (Big Sky)	Spring	Base, water equivalent	60	17.2	15.6	-1.7 in	-10%	15%
Whitefish & Flathead Range	Midwinter	Base, snow depth	25	49.9	45.1	-4.8 in	-10%	30%

SKI REGION	WINDOW	MEASURE	WINTERS	NORMAL	EL NIÑO	DIFFERENCE	CHANGE	R²
S Sangre de Cristo (Santa Fe)	Early season	Total new snow	46	24.8	31.7	+6.9 in	+28%	22%
Jemez (Pajarito)	Early season	Total new snow	46	20.0	25.7	+5.7 in	+28%	23%
S Sangre de Cristo (Santa Fe)	Spring	Base, water equivalent	46	9.5	12.3	+2.8 in	+29%	22%
Jemez (Pajarito)	Spring	Base, water equivalent	46	7.3	9.7	+2.4 in	+33%	23%
White Mtns AZ (Sunrise Park)	Spring	Base, water equivalent	46	4.3	6.3	+2.0 in	+46%	24%
S Sangre de Cristo (Santa Fe)	Early season	Base, snow depth	25	7.0	10.2	+3.2 in	+46%	32%



Only the results that survive false-discovery control, in inches. Three separate axes: inches of falling snow, inches of standing snow depth and inches of water in that snow are different quantities and cannot share a scale. The open circle is a normal winter; the filled circle is an El Niño winter.

RESULT

What survives, in days

Days within the window. A powder day is a day with at least six inches of new snow; a storm day at least two.								
SKI REGION	WINDOW	MEASURE	WINTERS	NORMAL	EL NIÑO	DIFFERENCE	CHANGE	R²
Missoula ranges (Montana Snowbowl)	Midwinter	Powder days (≥6 in)	46	1.4	0.8	-0.6 d	-42%	31%
Bighorn Mtns (Meadowlark)	Holidays	Storm days (≥2 in)	48	1.7	1.0	-0.7 d	-40%	20%
Park Range (Steamboat)	Holidays	Storm days (≥2 in)	47	4.3	2.7	-1.6 d	-36%	22%
Coeur d'Alenes (Silver/Lookout)	Midwinter	Powder days (≥6 in)	44	2.0	1.3	-0.7 d	-34%	22%
Snowy Range & Sierra Madre	Holidays	Storm days (≥2 in)	48	4.2	2.9	-1.3 d	-32%	17%
Central Cascades (Stevens/Snoqualmie)	Spring	Storm days (≥2 in)	45	8.5	6.4	-2.1 d	-25%	24%
Bitterroots (Lost Trail)	Midwinter	Storm days (≥2 in)	59	9.9	7.9	-2.0 d	-20%	13%
S Washington Cascades (Crystal/White Pass)	Spring	Storm days (≥2 in)	45	9.9	8.1	-1.8 d	-18%	16%
Madison & Gallatin (Big Sky)	Midwinter	Days with a skiable base	60	32.0	26.9	-5.1 d	-16%	13%
Missoula ranges (Montana Snowbowl)	Midwinter	Days with a skiable base	46	28.9	24.9	-4.0 d	-14%	25%
Whitefish & Flathead Range	Spring	Days with a skiable base	52	38.1	33.2	-4.9 d	-13%	18%
Missoula ranges (Montana Snowbowl)	Spring	Days with a skiable base	46	32.6	28.5	-4.1 d	-13%	33%
Wind River Range	Spring	Powder days (≥6 in)	48	0.6	0.8	+0.1 d	+21%	19%
White Mtns AZ (Sunrise Park)	Spring	Storm days (≥2 in)	46	2.3	2.9	+0.6 d	+24%	17%
S Sangre de Cristo (Santa Fe)	Early season	Storm days (≥2 in)	46	3.9	4.9	+1.0 d	+26%	19%
Jemez (Pajarito)	Early season	Storm days (≥2 in)	46	3.4	4.3	+0.9 d	+28%	20%
Chugach (Alyeska)	Holidays	Storm days (≥2 in)	43	4.5	5.7	+1.2 d	+28%	17%
S Sangre de Cristo (Santa Fe)	Spring	Days with a skiable base	46	22.1	33.3	+11.2 d	+51%	29%
Jemez (Pajarito)	Spring	Days with a skiable base	46	15.2	23.9	+8.7 d	+57%	21%
White Mtns AZ (Sunrise Park)	Spring	Days with a skiable base	46	8.2	14.3	+6.1 d	+74%	18%

What El Niño changes, in days — only results that survive false-discovery control



The same surviving results counted in days rather than inches.

READING THE LARGEST EFFECTS

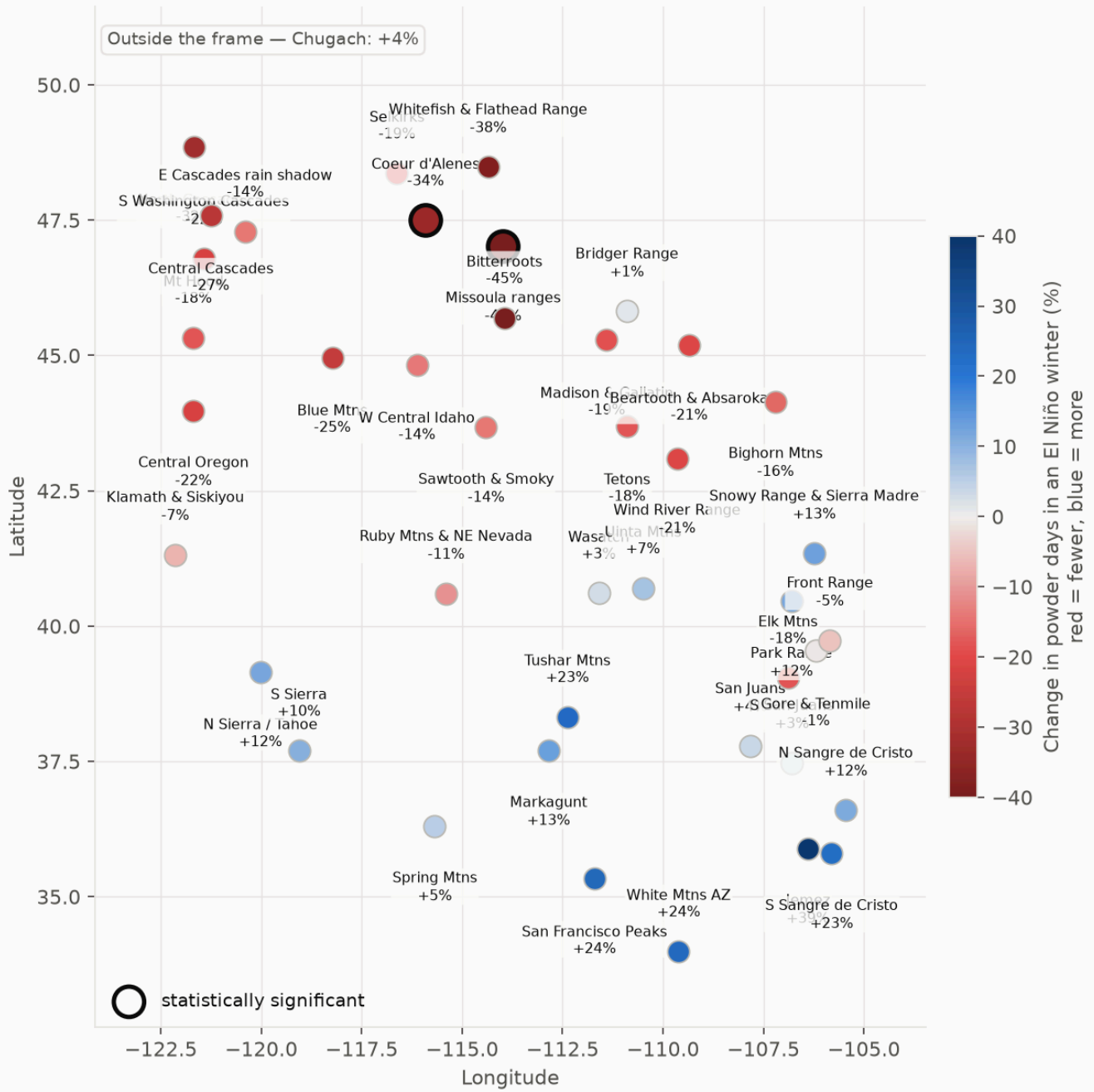
Montana Snowbowl in midwinter is the single most affected place in the study: twelve inches less new snow, two fewer storm days, and 45 % fewer powder days. The largest gain anywhere is northern New Mexico in spring, which picks up nearly nine additional days with a skiable base.

GEOGRAPHY

Where the effect lives

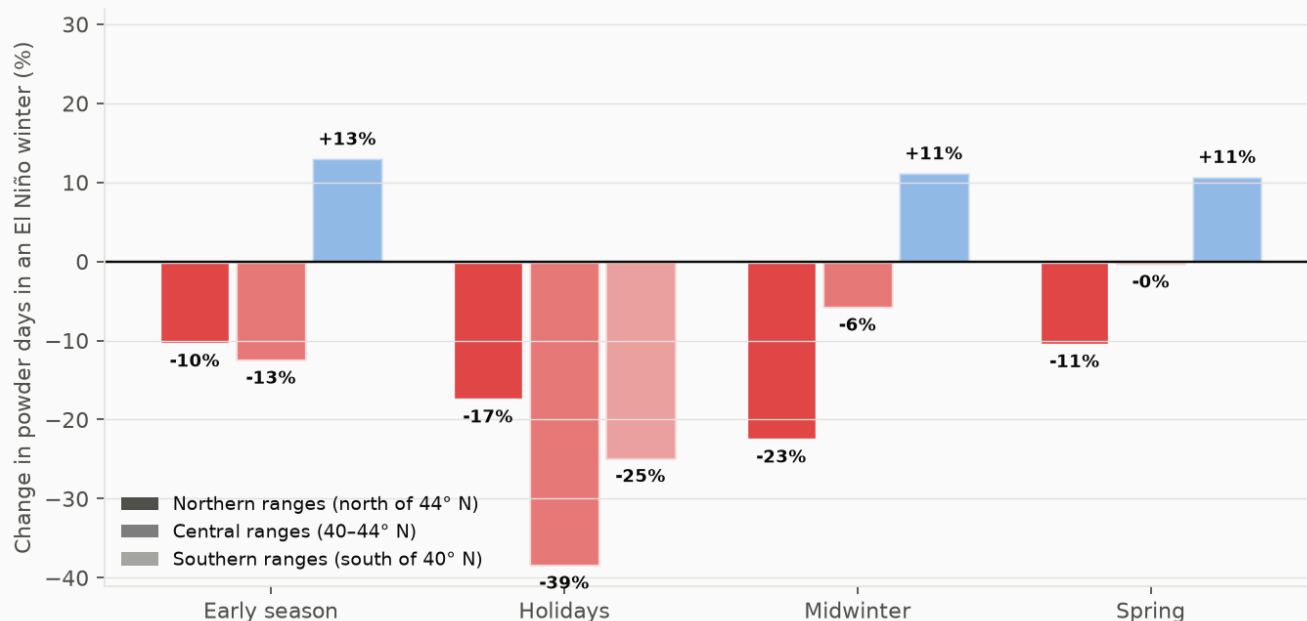
The turnover sits near 40–42° N, running through central Colorado and Utah. North of it El Niño suppresses snowfall; south of it, in spring, it adds. The southern half of the seesaw is weaker and less reliable than the northern half — several southern regions show a positive tendency that never reaches significance.

Where El Niño costs powder days — midwinter



Change in midwinter powder days by region. Ringed markers survive false-discovery control.

El Niño's effect on powder days, by window of winter and latitude

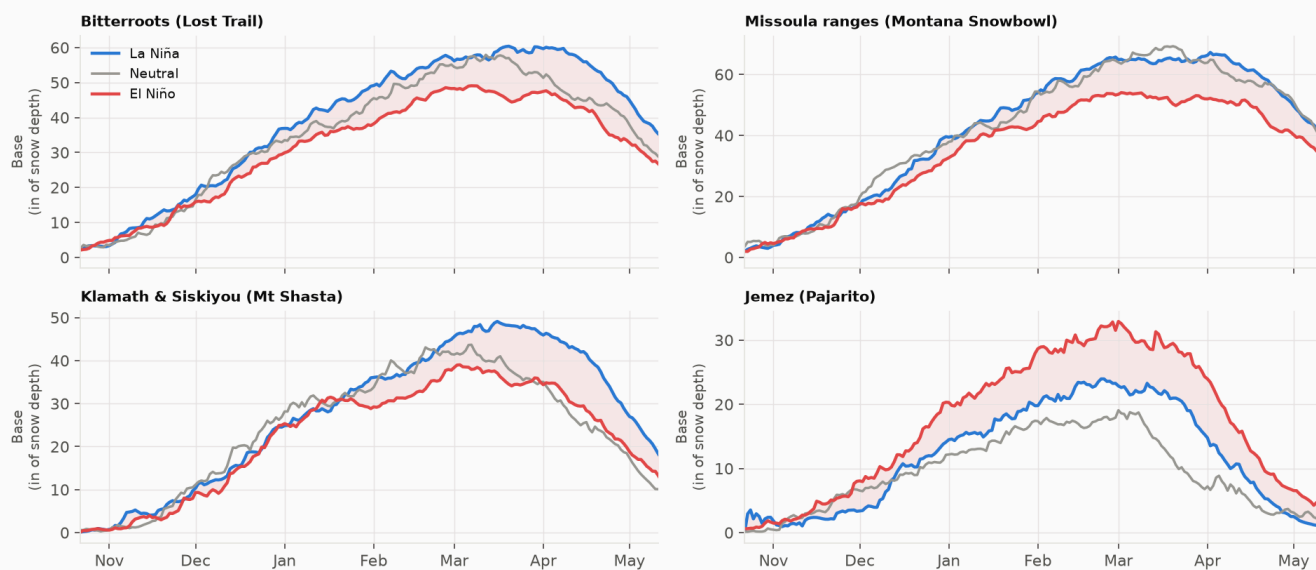


Change in powder days by window of winter, split at the ENSO node. Averaging all regions together would cancel the seesaw to near zero and say nothing true about anywhere; splitting makes both halves visible — and exposes the holiday window, which is negative in all three bands at once.

Direction consistency by window. Each test is one region and one measure; the sign test asks whether the direction holds across independent regions, which is the question that matters when each individual correlation is modest.

WINDOW OF WINTER	TESTS	LEANING TO LESS SNOW	SHARE	MEAN R	SIGN-TEST P
Early season	123	81 of 123	66%	-0.035	5.6e-04
Holidays	123	103 of 123	84%	-0.129	1.2e-14
Midwinter	123	89 of 123	72%	-0.098	7.4e-07
Spring	123	71 of 123	58%	-0.047	1.0e-01

How the base builds through the winter, by ENSO phase worst-hit regions at the top, a region that gains at the bottom right



Base depth through the winter by ENSO phase, in inches of measured snow depth. Two worst-hit regions at the top, a near-neutral region and a gaining region below.

GEOGRAPHY WITH UNCERTAINTY

The effect moves through the winter

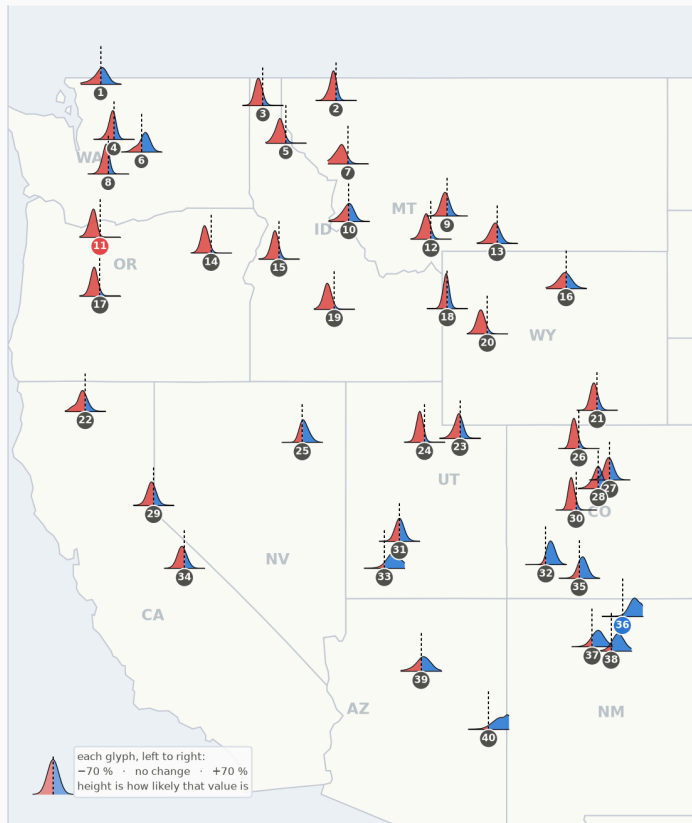
A single number per region hides how firmly it is pinned, so every ski region here carries its whole probability distribution rather than a colour. The dashed hairline through each glyph is no change. A density astride that line is a region where El Niño does nothing measurable.

Drawn window by window, the maps show what a season-long average cannot: **the effect does not sit still**. It is nearly absent in the early season, seizes the interior ranges over the holidays, moves to the northern Rockies and Cascades in midwinter, and retreats to the Cascades alone by spring. A region can be firmly hit in one window and indistinguishable from zero in the next.

Early season · 1 November – 15 December

Almost nothing is decided. Whatever El Niño does to a season has barely begun by mid-December.

The change in powder days in every ski region, with its uncertainty — early season



Ski regions, north to south

change in powder days · n.s. = interval includes zero

1 North Cascades	+0% n.s.	22 Klamath & Siskiyou	-11% n.s.
2 Whitefish & Flathead Range	-11% n.s.	23 Uinta Mtns	-6% n.s.
3 Selkirk	-15% n.s.	24 Wasatch	-17% n.s.
4 Central Cascades	-6% n.s.	25 Ruby Mtns & NE Nevada	+7% n.s.
5 Coeur d'Alenes	-23% n.s.	26 Park Range	-16% n.s.
6 E Cascades rain shadow	+10% n.s.	27 Front Range	-2% n.s.
7 Missoula ranges	-28% n.s.	28 Gore & Tenmile	-3% n.s.
8 S Washington Cascades	-9% n.s.	29 N Sierra / Tahoe	-7% n.s.
9 Bridger Range	-7% n.s.	30 Elk Mtns	-16% n.s.
10 Bitterroots	-5% n.s.	31 Tushar Mtns	+1% n.s.
11 Mt Hood	-26%	32 San Juans	+19% n.s.
12 Madison & Gallatin	-18% n.s.	33 Markagunt	+33% n.s.
13 Beartooth & Absaroka	-8% n.s.	34 S Sierra	-8% n.s.
14 Blue Mtns	-22% n.s.	35 S San Juans	+12% n.s.
15 W Central Idaho	-15% n.s.	36 N Sangre de Cristo	+44%
16 Bighorn Mtns	-5% n.s.	37 Jemez	+20% n.s.
17 Central Oregon	-21% n.s.	38 S Sangre de Cristo	+26% n.s.
18 Teton	-4% n.s.	39 San Francisco Peaks	+7% n.s.
19 Sawtooth & Smoky	-25% n.s.	40 White Mtns AZ	+46% n.s.
20 Wind River Range	-23% n.s.	Chugach (not on map)	+11% n.s.
21 Snowy Range & Sierra Madre	-11% n.s.		

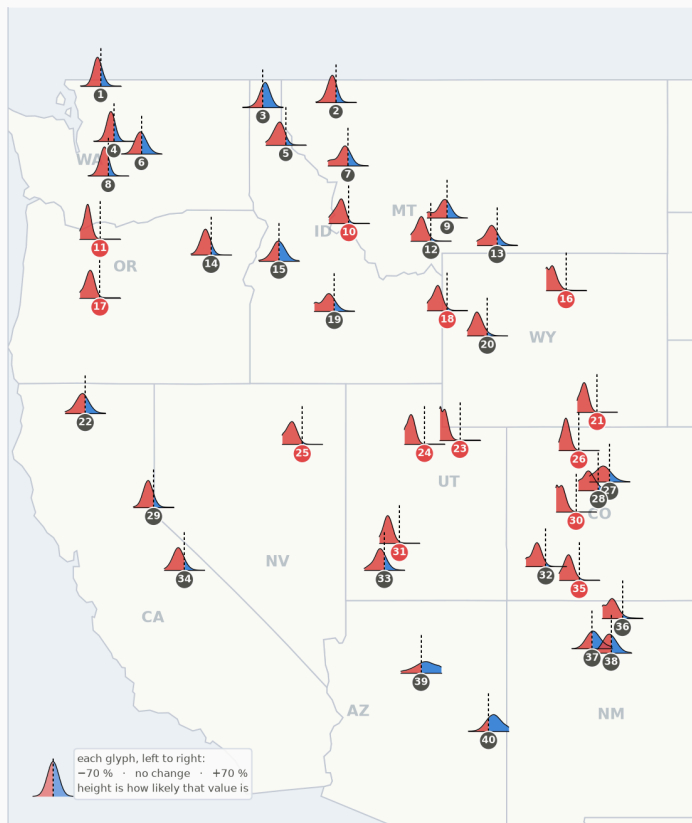
Each glyph is a probability density from 10,000 resamples of the record, calibrated so its width is an honest sampling distribution. 38 of 40 regions straddle zero and are marked n.s.

2 of 41 regions clear zero.

Holidays · 16 December – 5 January

The widest, most coherent signal of the winter, and it lands on the interior: the Wasatch, the Uintas, the Elk Mountains, the Park and Snowy ranges, the Rubies. These are the same regions that show nothing in midwinter.

The change in powder days in every ski region, with its uncertainty — holidays



Ski regions, north to south

change in powder days · n.s. = interval includes zero

1 North Cascades	-9% n.s.	22 Klamath & Siskiyou	-11% n.s.
2 Whitefish & Flathead Range	-14% n.s.	23 Uinta Mtns	-58%
3 Selkirk	+7% n.s.	24 Wasatch	-49%
4 Central Cascades	-11% n.s.	25 Ruby Mtns & NE Nevada	-40%
5 Coeur d'Alenes	-25% n.s.	26 Park Range	-45%
6 E Cascades rain shadow	-1% n.s.	27 Front Range	-25% n.s.
7 Missoula ranges	-17% n.s.	28 Gore & Tenmile	-36% n.s.
8 S Washington Cascades	-14% n.s.	29 N Sierra / Tahoe	-19% n.s.
9 Bridger Range	-16% n.s.	30 Elk Mtns	-54%
10 Bitterroots	-34%	31 Tushar Mtns	-40%
11 Mt Hood	-44%	32 San Juans	-35% n.s.
12 Madison & Gallatin	-36% n.s.	33 Markagunt	-16% n.s.
13 Beartooth & Absaroka	-21% n.s.	34 S Sierra	-22% n.s.
14 Blue Mtns	-21% n.s.	35 S San Juans	-39%
15 W Central Idaho	-2% n.s.	36 N Sangre de Cristo	-41% n.s.
16 Bighorn Mtns	-58%	37 Jemez	+7% n.s.
17 Central Oregon	-37%	38 S Sangre de Cristo	-5% n.s.
18 Tetons	-37%	39 San Francisco Peaks	+16% n.s.
19 Sawtooth & Smoky	-27% n.s.	40 White Mtns AZ	+20% n.s.
20 Wind River Range	-34% n.s.	Chugach (not on map)	+20% n.s.
21 Snowy Range & Sierra Madre	-47%		

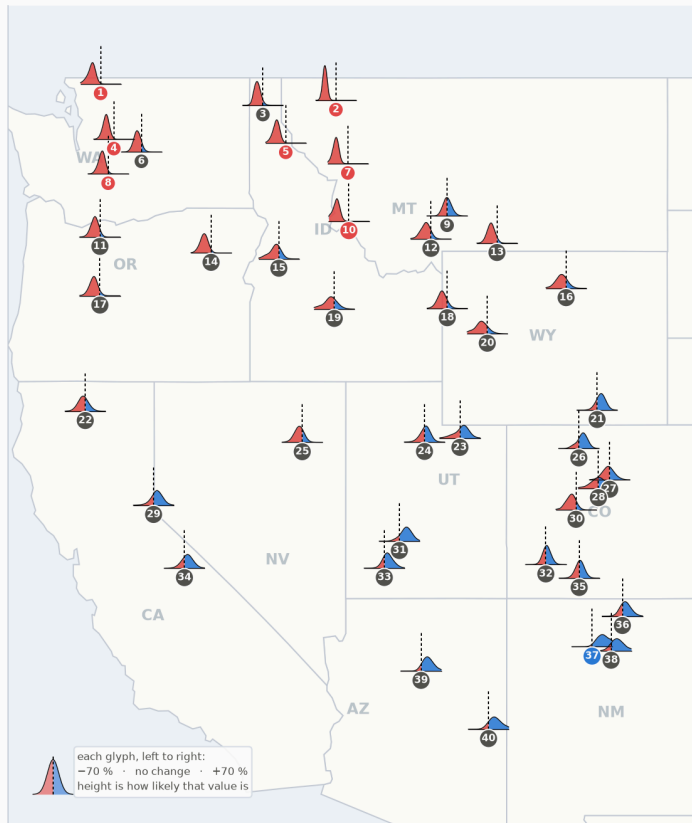
Each glyph is a probability density from 10,000 resamples of the record, calibrated so its width is an honest sampling distribution. 27 of 40 regions straddle zero and are marked n.s.

13 of 41 regions clear zero, and the cluster sits over the Great Basin and the Colorado and Wyoming ranges rather than the north.

Midwinter · 6 January – 28 February

The signal migrates north, to the Bitterroots, the Missoula ranges, the Coeur d'Alenes, Whitefish and the Washington Cascades, with northern New Mexico gaining. The interior ranges that were firmly hit over the holidays are now undecided.

The change in powder days in every ski region, with its uncertainty — midwinter



Ski regions, north to south

change in powder days · n.s. = interval includes zero

1 North Cascades	-32%	22 Klamath & Siskiyou	-7% n.s.
2 Whitefish & Flathead Range	-38%	23 Uinta Mtns	+7% n.s.
3 Selkirk	-19% n.s.	24 Wasatch	+3% n.s.
4 Central Cascades	-27%	25 Ruby Mtns & NE Nevada	-11% n.s.
5 Coeur d'Alenes	-34%	26 Park Range	+12% n.s.
6 E Cascades rain shadow	-14% n.s.	27 Front Range	-5% n.s.
7 Missoula ranges	-42%	28 Gore & Tenmile	-1% n.s.
8 S Washington Cascades	-22%	29 N Sierra / Tahoe	+12% n.s.
9 Bridger Range	+1% n.s.	30 Elk Mtns	-18% n.s.
10 Bitterroots	-45%	31 Tushar Mtns	+23% n.s.
11 Mt Hood	-18% n.s.	32 San Juans	+4% n.s.
12 Madison & Gallatin	-19% n.s.	33 Markagunt	+13% n.s.
13 Beartooth & Absaroka	-21% n.s.	34 S Sierra	+10% n.s.
14 Blue Mtns	-25% n.s.	35 S San Juans	+3% n.s.
15 W Central Idaho	-14% n.s.	36 N Sangre de Cristo	+12% n.s.
16 Bighorn Mtns	-16% n.s.	37 Jemez	+39%
17 Central Oregon	-22% n.s.	38 S Sangre de Cristo	+23% n.s.
18 Tetons	-18% n.s.	39 San Francisco Peaks	+24% n.s.
19 Sawtooth & Smoky	-14% n.s.	40 White Mtns AZ	+24% n.s.
20 Wind River Range	-21% n.s.	Chugach (not on map)	+4% n.s.
21 Snowy Range & Sierra Madre	+13% n.s.		

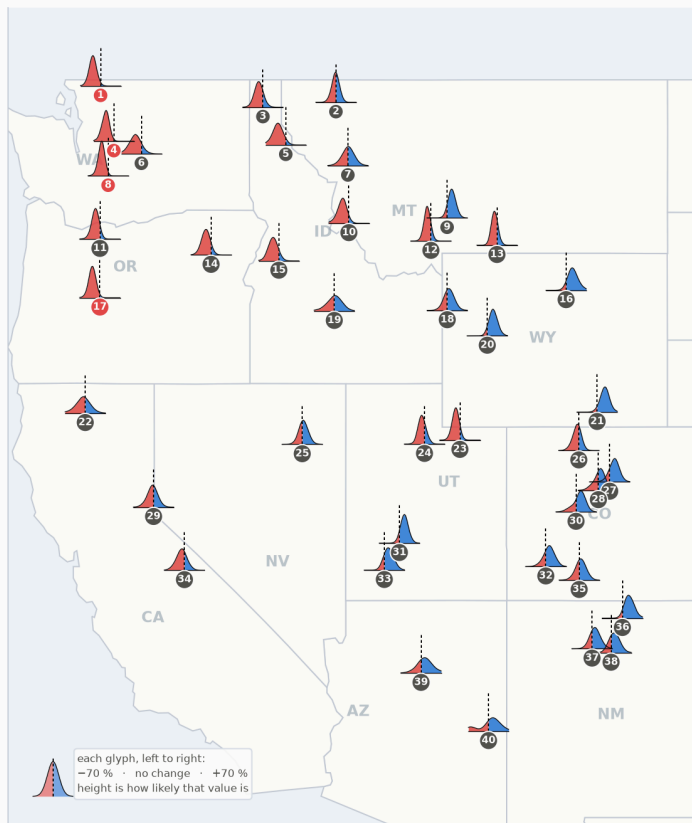
Each glyph is a probability density from 10,000 resamples of the record, calibrated so its width is an honest sampling distribution. 32 of 40 regions straddle zero and are marked n.s.

8 of 41 regions clear zero.

Spring · 1 March – 15 April

Only the Cascades and central Oregon remain decided.

The change in powder days in every ski region, with its uncertainty — spring



Ski regions, north to south

change in powder days · n.s. = interval includes zero

1 North Cascades	-27%	22 Klamath & Siskiyou	-8% n.s.
2 Whitefish & Flathead Range	-1% n.s.	23 Uinta Mtns	-15% n.s.
3 Selkirk	-14% n.s.	24 Wasatch	-8% n.s.
4 Central Cascades	-29%	25 Ruby Mtns & NE Nevada	+4% n.s.
5 Coeur d'Alenes	-28% n.s.	26 Park Range	-7% n.s.
6 E Cascades rain shadow	-22% n.s.	27 Front Range	+16% n.s.
7 Missoula ranges	+1% n.s.	28 Gore & Tenmile	+4% n.s.
8 S Washington Cascades	-23%	29 N Sierra / Tahoe	-4% n.s.
9 Bridger Range	+17% n.s.	30 Elk Mtns	+12% n.s.
10 Bitterroots	-24% n.s.	31 Tushar Mtns	+18% n.s.
11 Mt Hood	-16% n.s.	32 San Juans	+12% n.s.
12 Madison & Gallatin	-13% n.s.	33 Markagunt	+16% n.s.
13 Beartooth & Absaroka	-8% n.s.	34 S Sierra	-10% n.s.
14 Blue Mtns	-19% n.s.	35 S San Juans	+6% n.s.
15 W Central Idaho	-20% n.s.	36 N Sangre de Cristo	+24% n.s.
16 Bighorn Mtns	+24% n.s.	37 Jemez	+11% n.s.
17 Central Oregon	-27%	38 S Sangre de Cristo	+12% n.s.
18 Tetons	+7% n.s.	39 San Francisco Peaks	+13% n.s.
19 Sawtooth & Smoky	+1% n.s.	40 White Mtns AZ	+5% n.s.
20 Wind River Range	+21% n.s.	Chugach (not on map)	+24% n.s.
21 Snowy Range & Sierra Madre	+27% n.s.		

Each glyph is a probability density from 10,000 resamples of the record, calibrated so its width is an honest sampling distribution. 36 of 40 regions straddle zero and are marked n.s.

4 of 41 regions clear zero, all in the Cascades and Oregon.

TWO STANDARDS OF EVIDENCE, AND WHY THEY DISAGREE

These maps judge each region on its own: does its 95 % interval exclude zero? The headline tables apply Benjamini–Hochberg correction across all 971 tests at once, which is far stricter, because asking a question of 41 regions in four windows on six measures will turn up apparent effects by chance alone.

That is why a region can appear here at -50 % over the holidays and not appear in the list of results that survive correction. The per-region interval answers "what does the record say about *this* range"; the corrected grid answers "which of these findings would I bet on". A region that clears the first but not the second is worth watching, not worth relying on.

Every region whose 95 % interval excludes zero, by window. Percentages of powder days sit next to the inches of new snow behind them, because a twenty-one-day window has few powder days anywhere and a large percentage can rest on a fraction of a day.

WINDOW	SKI REGION	POWDER DAYS, NORMAL	CHANGE	NEW SNOW, NORMAL	EL NIÑO	DIFFERENCE	95 % INTERVAL

WINDOW	SKI REGION	POWDER DAYS, NORMAL	CHANGE	NEW SNOW, NORMAL	EL NIÑO	DIFFERENCE	95 % INTERVAL
Early season	Mt Hood	2.30	-26%	51 in	42 in	-10 in	-54% to -4%
	N Sangre de Cristo	0.62	+44%	26 in	32 in	+6 in	+7% to +84%
Holidays	Bighorn Mtns TINY BASE	0.10	-58%	12 in	8 in	-4 in	-120% to -14%
	Uinta Mtns	0.54	-58%	17 in	11 in	-6 in	-90% to -30%
	Elk Mtns	0.57	-54%	20 in	15 in	-6 in	-89% to -21%
	Wasatch	0.99	-49%	23 in	15 in	-8 in	-80% to -21%
	Snowy Range & Sierra Madre	0.79	-47%	25 in	18 in	-7 in	-78% to -19%
	Park Range	0.85	-45%	25 in	17 in	-8 in	-73% to -18%
	Mt Hood	1.53	-44%	33 in	26 in	-7 in	-72% to -19%
	Tushar Mtns	0.69	-40%	18 in	16 in	-2 in	-72% to -8%
	Ruby Mtns & NE Nevada	0.65	-40%	19 in	14 in	-5 in	-81% to -4%
	S San Juans	1.03	-39%	22 in	18 in	-4 in	-74% to -4%
	Tetons	0.91	-37%	26 in	20 in	-7 in	-77% to -3%
	Central Oregon	1.56	-37%	35 in	28 in	-6 in	-72% to -8%
	Bitterroots	0.73	-34%	24 in	20 in	-4 in	-76% to -3%
	Bitterroots	1.60	-45%	60 in	48 in	-12 in	-71% to -22%
	Missoula ranges	1.45	-42%	57 in	47 in	-10 in	-64% to -24%
Midwinter	Whitefish & Flathead Range	1.55	-38%	59 in	48 in	-11 in	-53% to -24%
	Coeur d'Alenes	2.02	-34%	63 in	53 in	-10 in	-59% to -12%
	North Cascades	4.63	-32%	103 in	85 in	-18 in	-62% to -9%
	Central Cascades	3.62	-27%	84 in	71 in	-13 in	-50% to -7%
	S Washington Cascades	3.92	-22%	89 in	79 in	-10 in	-46% to -0%
	Jemez	0.73	+39%	30 in	37 in	+7 in	+1% to +84%
	Central Cascades	1.71	-29%	50 in	39 in	-12 in	-56% to -6%
Spring	North Cascades	3.19	-27%	79 in	67 in	-12 in	-53% to -2%
	Central Oregon	1.90	-27%	54 in	46 in	-9 in	-51% to -3%
	S Washington Cascades	2.08	-23%	60 in	50 in	-11 in	-47% to -2%

THE NEGATIVE RESULT

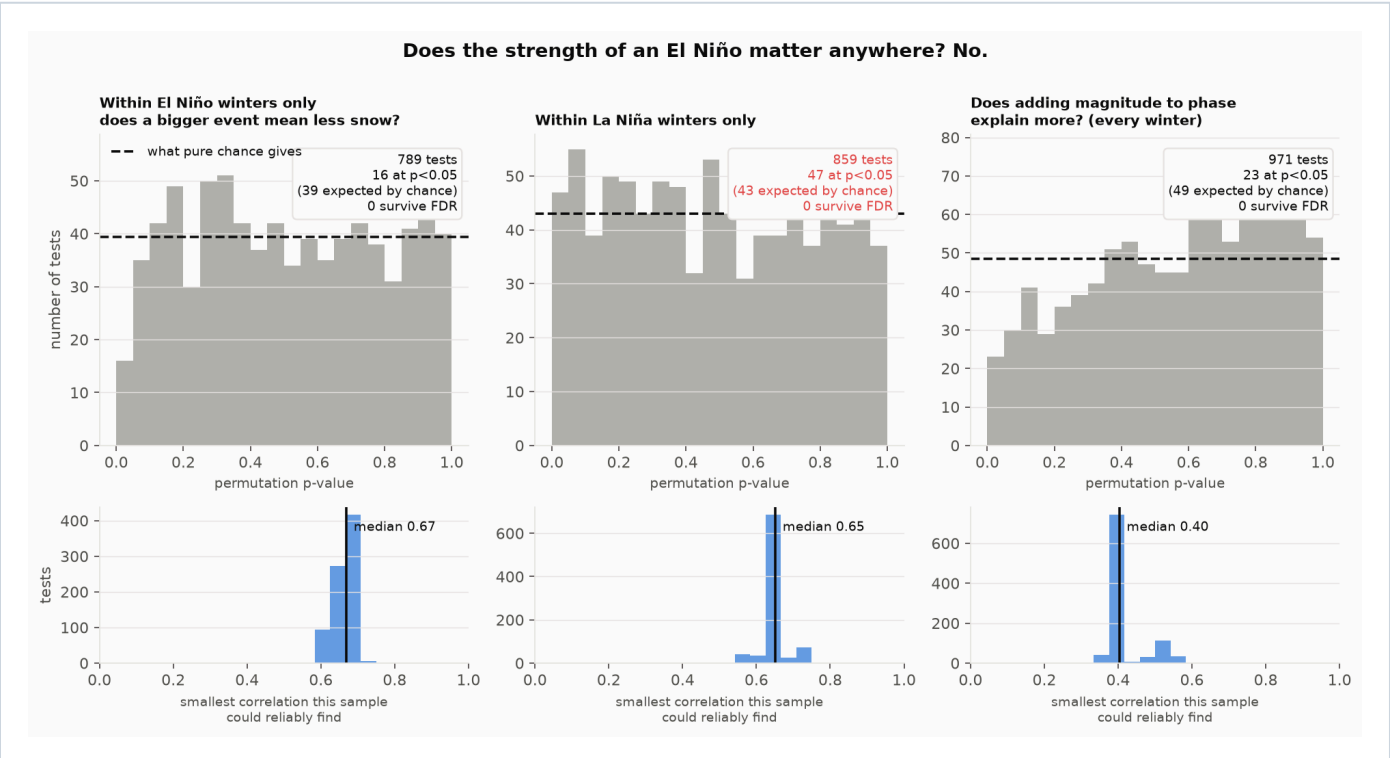
Event strength carries no information

Phase says which way a winter leans. Strength says how far the index went. The two are routinely conflated in seasonal forecasting, so strength was tested head-on rather than inferred: every region, window and measure was asked whether a larger event means less snow among El Niño winters only, the same among La Niña winters, and whether adding the index value to a phase-only model explains more.

2619 strength tests with permutation p-values and false-discovery control across the whole family.					
TEST	TESTS RUN	P < 0.05	EXPECTED BY CHANCE	SURVIVE FDR	SMALLEST R FINDABLE
Within El Niño winters only	789	16	39	0	0.67
Within La Niña winters only	859	47	43	0	0.65

TEST	TESTS RUN	P < 0.05	EXPECTED BY CHANCE	SURVIVE FDR	SMALLEST R FINDABLE
Adding magnitude to phase (every winter)	971	23	49	0	0.40

Nothing survives. The p-values are distributed exactly as pure chance distributes them, and the within-El Niño family returns *fewer* nominal hits than chance alone would produce.



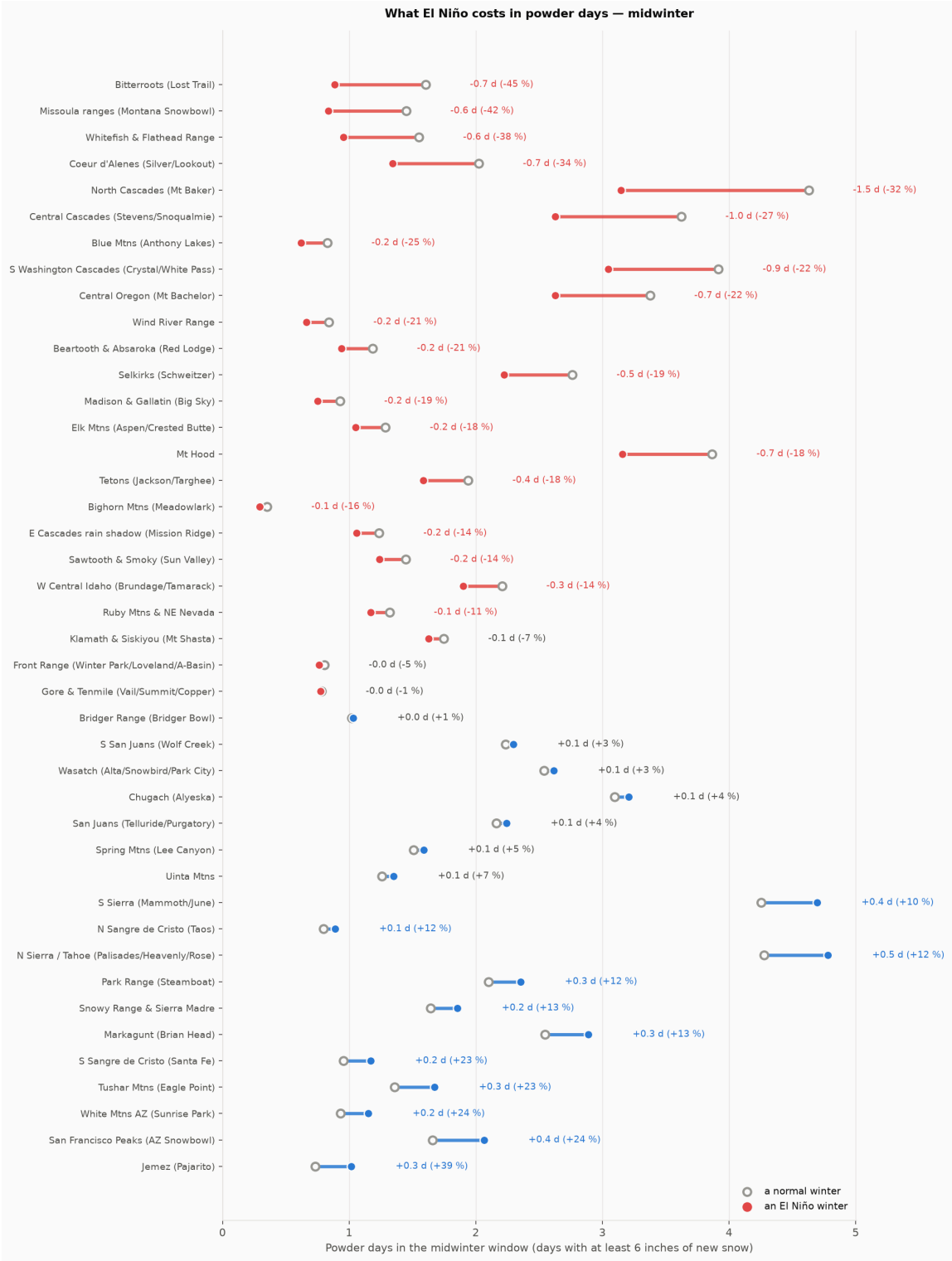
Permutation p-values for every strength test against the uniform expectation. A real signal would pile up at the left edge. The lower row is the power check: with about sixteen winters of a given phase per region, a within-phase test could only have found a correlation above roughly 0.65, so its flat histogram means not found rather than not there. The nested test uses every winter, could have found 0.40, and did not.

Midwinter powder days by strength bin, averaged over every region. If magnitude mattered the bars would descend from left to right.

PHASE AND STRENGTH	WINTERS	POWDER DAYS
La Niña · strong	6	1.63
La Niña · moderate	5	1.31
La Niña · weak	10	1.37
Neutral · neutral	18	1.33
El Niño · weak	9	1.23
El Niño · moderate	4	1.18
El Niño · strong	4	1.09
El Niño · very strong	3	1.40

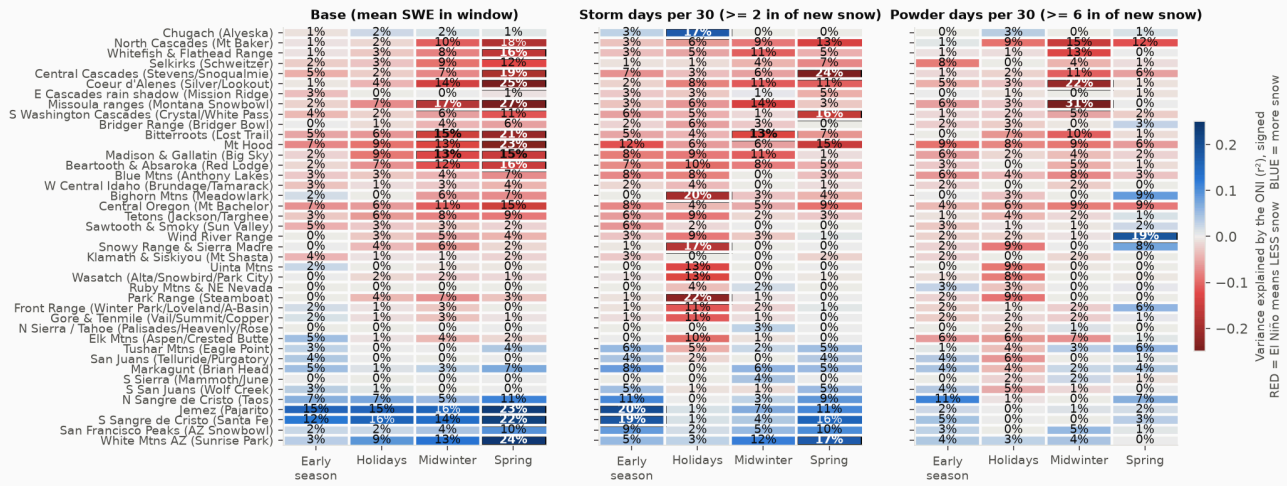
Ski Region	States	Snow Climate	Winters	Normal	El Niño	Diff	Change	La Niña	Status
Bitterroots (Lost Trail)	MT+ID	transitional	59	1.6	0.9	-0.7	-45%	2.3	SURVIVES FDR
Missoula ranges (Montana Snowbowl)	MT	transitional	46	1.4	0.8	-0.6	-42%	2.0	SURVIVES FDR
Whitefish & Flathead Range	MT	transitional	52	1.5	1.0	-0.6	-38%	2.0	SURVIVES FDR
Coeur d'Alenes (Silver/Lookout)	ID+MT	transitional	44	2.0	1.3	-0.7	-34%	2.7	SURVIVES FDR
North Cascades (Mt Baker)	WA	maritime	31	4.6	3.1	-1.5	-32%	5.8	NOT SIGNIFICANT
Central Cascades (Stevens/Snoqualmie)	WA	maritime	45	3.6	2.6	-1.0	-27%	4.3	SURVIVES FDR
Blue Mtns (Anthony Lakes)	OR	transitional	48	0.8	0.6	-0.2	-25%	0.9	NOT SIGNIFICANT
S Washington Cascades (Crystal/White Pass)	WA	maritime	45	3.9	3.0	-0.9	-22%	4.3	SURVIVES FDR
Central Oregon (Mt Bachelor)	OR	transitional	46	3.4	2.6	-0.7	-22%	4.2	NOT SIGNIFICANT
Wind River Range	WY	continental	48	0.8	0.7	-0.2	-21%	0.7	SURVIVES FDR
Beartooth & Absaroka (Red Lodge)	MT+WY	continental	53	1.2	0.9	-0.2	-21%	1.5	SURVIVES FDR
Selkirks (Schweitzer)	ID	transitional	43	2.8	2.2	-0.5	-19%	3.3	SURVIVES FDR
Madison & Gallatin (Big Sky)	MT	continental	60	0.9	0.8	-0.2	-19%	1.0	SURVIVES FDR
Elk Mtns (Aspen/Crested Butte)	CO	continental	46	1.3	1.1	-0.2	-18%	1.5	NOT SIGNIFICANT
Mt Hood	OR	maritime	48	3.9	3.2	-0.7	-18%	4.7	SURVIVES FDR
Tetons (Jackson/Targhee)	WY+ID	intermountain	46	1.9	1.6	-0.4	-18%	1.8	NOT SIGNIFICANT
Bighorn Mtns (Meadowlark)	WY	continental	48	0.4	0.3	-0.1	-16%	0.4	SURVIVES FDR
E Cascades rain shadow (Mission Ridge)	WA	transitional	45	1.2	1.1	-0.2	-14%	1.5	NOT SIGNIFICANT
Sawtooth & Smoky (Sun Valley)	ID	intermountain	46	1.4	1.2	-0.2	-14%	1.3	NOT SIGNIFICANT
W Central Idaho (Brundage/Tamarack)	ID	intermountain	46	2.2	1.9	-0.3	-14%	2.3	NOT SIGNIFICANT
Ruby Mtns & NE Nevada	NV	continental	46	1.3	1.2	-0.1	-11%	1.4	NOT SIGNIFICANT
Klamath & Siskiyou (Mt Shasta)	CA+OR	maritime	46	1.7	1.6	-0.1	-7%	1.8	NOT SIGNIFICANT
Front Range (Winter Park/Loveland/A-Basin)	CO	continental	48	0.8	0.8	-0.0	-5%	0.8	NOT SIGNIFICANT
Gore & Tenmile (Vail/Summit/Copper)	CO	continental	48	0.8	0.8	-0.0	-1%	0.8	NOT SIGNIFICANT
Bridger Range (Bridger Bowl)	MT	continental	50	1.0	1.0	+0.0	+1%	1.0	NOT SIGNIFICANT
S San Juans (Wolf Creek)	CO	continental	46	2.2	2.3	+0.1	+3%	2.4	NOT SIGNIFICANT
Wasatch (Alta/Snowbird/Park City)	UT	intermountain	48	2.5	2.6	+0.1	+3%	2.5	NOT SIGNIFICANT
Chugach (Alyeska)	AK	maritime	43	3.1	3.2	+0.1	+4%	3.1	SURVIVES FDR
San Juans (Telluride/Purgatory)	CO	continental	46	2.2	2.2	+0.1	+4%	2.2	NOT SIGNIFICANT
Spring Mtns (Lee Canyon)	NV	continental	18	1.5	1.6	+0.1	+5%	1.3	NOT SIGNIFICANT
Uinta Mtns	UT	continental	48	1.3	1.4	+0.1	+7%	1.1	NOT SIGNIFICANT
S Sierra (Mammoth/June)	CA	maritime	48	4.3	4.7	+0.4	+10%	4.0	NOT SIGNIFICANT
N Sangre de Cristo (Taos)	NM+CO	continental	47	0.8	0.9	+0.1	+12%	0.7	NOT SIGNIFICANT
N Sierra / Tahoe (Palisades/Heavenly/Rose)	CA+NV	maritime	48	4.3	4.8	+0.5	+12%	4.1	NOT SIGNIFICANT
Park Range (Steamboat)	CO	continental	47	2.1	2.4	+0.3	+12%	2.1	SURVIVES FDR
Snowy Range & Sierra Madre	WY	continental	48	1.6	1.9	+0.2	+13%	1.6	SURVIVES FDR
Markagunt (Brian Head)	UT	continental	46	2.5	2.9	+0.3	+13%	2.3	NOT SIGNIFICANT
S Sangre de Cristo (Santa Fe)	NM	continental	46	1.0	1.2	+0.2	+23%	0.9	SURVIVES FDR
Tushar Mtns (Eagle Point)	UT	continental	47	1.4	1.7	+0.3	+23%	1.3	NOT SIGNIFICANT

SKI REGION	STATES	SNOW CLIMATE	WINTERS	NORMAL	EL NIÑO	DIFF	CHANGE	LA NIÑA	STATUS
White Mtns AZ (Sunrise Park)	AZ	continental	46	0.9	1.2	+0.2	+24%	0.6	SURVIVES FDR
San Francisco Peaks (AZ Snowbowl)	AZ	continental	46	1.7	2.1	+0.4	+24%	1.4	NOT SIGNIFICANT
Jemez (Pajarito)	NM	continental	46	0.7	1.0	+0.3	+39%	0.7	SURVIVES FDR



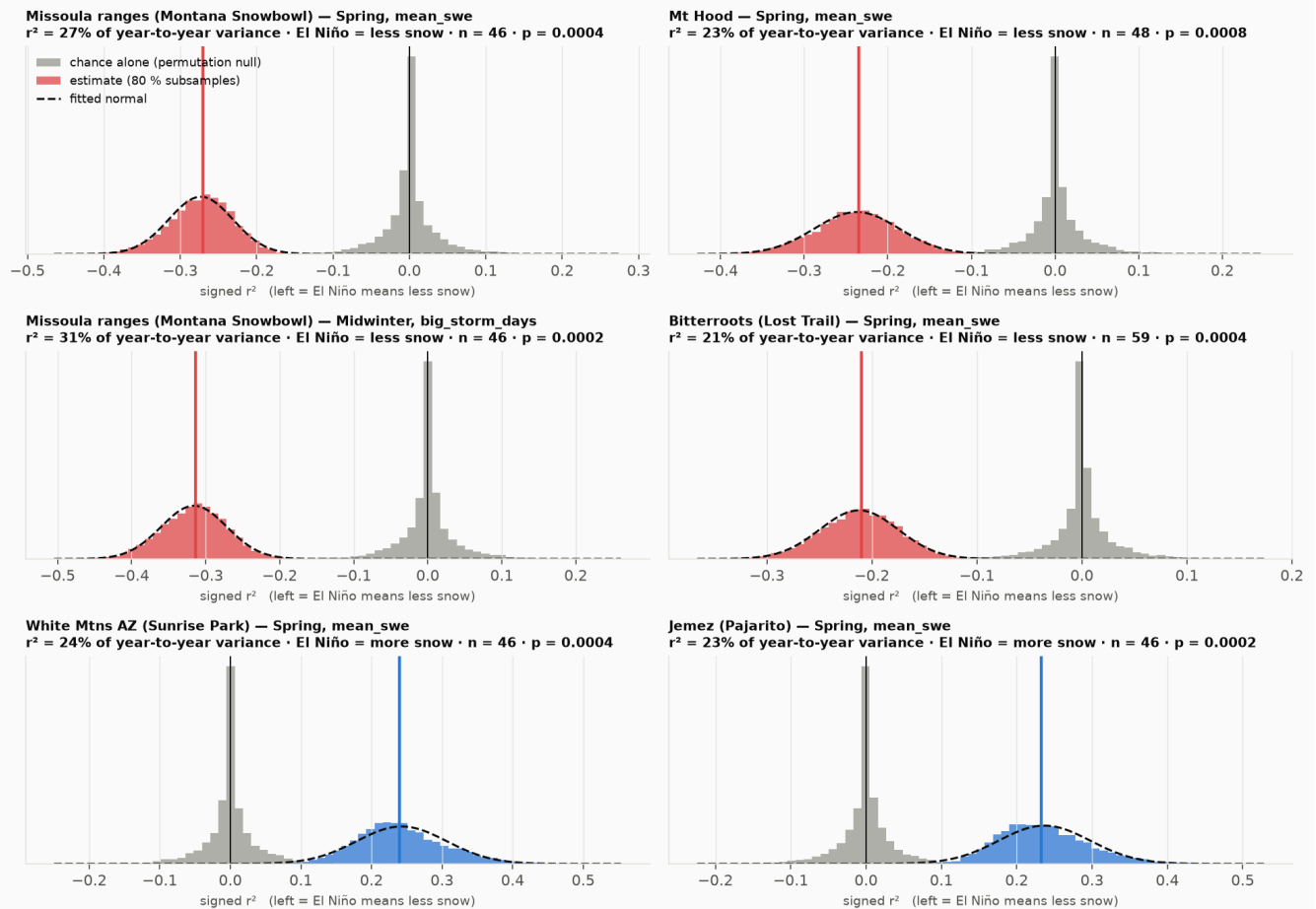
Powder days in a normal winter against an El Niño winter, every region with data, sorted by effect.

El Niño and the ski season, by region and window of winter

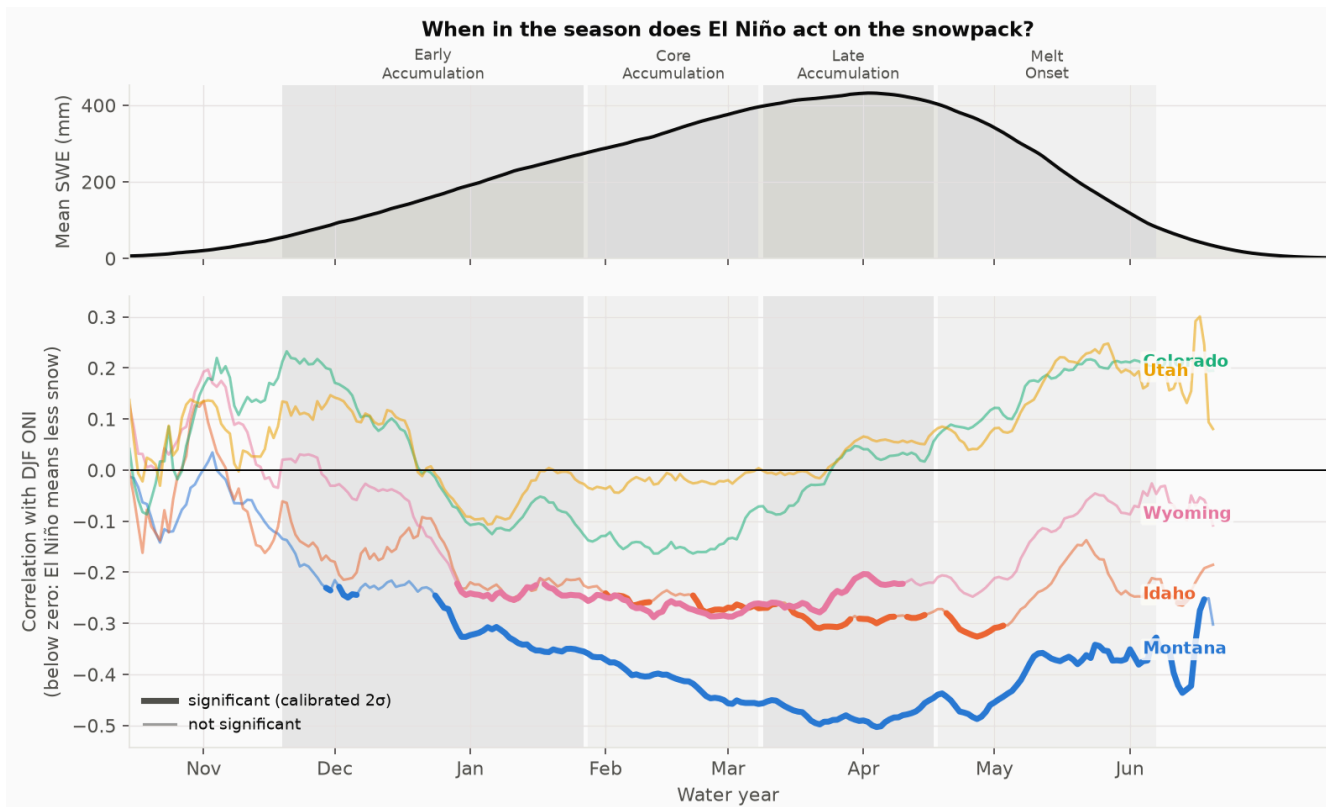


Variance explained (r^2) by region, window and measure, regions ordered north to south. The vertical gradient is the seesaw.

Probability distributions behind each result



Probability distributions behind selected headline results: the subsampling distribution of the estimate against the permutation null, with the fitted normal. Separation between the two is the significance, shown rather than asserted.



The correlation with the DJF index for every day of the water year, against the snowpack climatology, showing when in the season the signal acts.

METHOD

How this was measured

A powder day is six inches of snow

A fixed water threshold is a maritime yardstick. Cascade new snow runs about 12 % density and Colorado's about 7 %, so 25 mm of water is roughly eight inches of Cascade snow but fourteen inches of Colorado snow — counting water alone makes the driest, lightest snowpacks look storm-free. Each station's own new-snow ratio is measured from the overlap of its depth and water records, then applied to its much longer water record. Across 508 stations the median ratio is 7.5 inches of snow per inch of water.

Regions are ranges, not states

A state boundary is not a snow boundary. Tahoe spans California and Nevada, the Bitterroots and Selkirks span Montana and Idaho, the Tetons span Wyoming and Idaho, the Sangre de Cristo spans Colorado and New Mexico. Each region is anchored on a mountain range around real ski terrain and labelled with its snow climate.

Stations are weighted by elevation

A gauge represents a ski area only insofar as it shares its snow climate, and elevation is most of that. Each region carries its served base-to-summit band; stations are weighted by a Gaussian on the vertical gap to that band and another on horizontal distance, so a valley gauge below the lifts does not speak for a mountain.

Significance, and the correction for asking often

The modified bootstrap of Brown & Harper (2026): omit a random 20 % of winters, refit, repeat ten thousand times, and require the 2σ bounds of the resulting near-Gaussian distribution to exclude zero. That raw rule fires on about 30 % of pure noise, because the subsample spread understates the sampling error by the delete-*d* factor $\sqrt{((n-d)/d)} = 2$, so a calibrated version and a permutation p-value are reported alongside. Benjamini-Hochberg control is then applied across the entire grid.

Two views of the same data

The statistical tests run on per-station detrended z-scores, so a long-term warming trend cannot pose as an ENSO signal. The physical composites are raw averages by phase — what actually happened. The two are matched explicitly on region, window and measure rather than assumed to line up.

Windows that carry a season

Early season (1 Nov – 15 Dec), Holidays (16 Dec – 5 Jan), Midwinter (6 Jan – 28 Feb) and Spring (1 Mar – 15 Apr). They abut without gaps. These replace the hydrologic periods used in water-supply work, which are built around runoff timing rather than revenue.

COVERAGE

Regions and station coverage

HOW A REGION'S LOCATION IS FIXED

Each region's centre is computed from the coordinates of the ski areas it serves, never typed by hand. An earlier draft did type them, and drifted: one region's centre sat 61 km from Schweitzer, and another grouped Lookout Pass with Montana Snowbowl 138 km away under a name matching neither. The regions are now split where their areas were too far apart to share a snow climate — the Selkirks from the Coeur d'Alenes, Bridger Bowl from Big Sky, Wolf Creek from Telluride, Taos from Santa Fe — and a test checks that every named ski area falls inside a state its region claims.

Effective station count is Kish's measure applied to the elevation weights. Regions with few stations inside the served band — notably southern Utah and the Snowy Range — should be read with more caution, because SNOTEL does not reach those elevations locally.

SKI REGION	STATES	CLIMATE	SERVED BAND (FT)	STATIONS	EFFECTIVE	WEIGHTED ELEV (FT)	IN BAND
Beartooth & Absaroka (Red Lodge)	MT+WY	continental	7,016–9,416	14	12.4	8,101	10
Bighorn Mtns (Meadowlark)	WY	continental	7,500–9,500	15	13.4	8,716	12
Bitterroots (Lost Trail)	MT+ID	transitional	6,400–8,000	13	10.8	7,000	5
Blue Mtns (Anthony Lakes)	OR	transitional	7,100–8,000	18	13.7	5,809	2
Bridger Range (Bridger Bowl)	MT	continental	6,100–8,700	6	5.2	7,156	6
Central Cascades (Stevens/Snoqualmie)	WA	maritime	3,000–5,845	24	20.9	3,832	20
Central Oregon (Mt Bachelor)	OR	transitional	6,300–9,065	10	7.3	5,220	0
Chugach (Alyeska)	AK	maritime	250–2,750	11	9.5	1,413	7
Coeur d'Alenes (Silver/Lookout)	ID+MT	transitional	4,100–6,300	8	6.3	5,055	7

SKI REGION	STATES	CLIMATE	SERVED BAND (FT)	STATIONS	EFFECTIVE	WEIGHTED ELEV (FT)	IN BAND
E Cascades rain shadow (Mission Ridge)	WA	transitional	4,570–6,820	7	5.9	4,744	2
Elk Mtns (Aspen/Crested Butte)	CO	continental	7,945–12,162	15	13.8	10,139	15
Front Range (Winter Park/Loveland/A-Basin)	CO	continental	9,000–13,050	26	22.2	10,403	22
Gore & Tenmile (Vail/Summit/Copper)	CO	continental	8,120–12,998	25	22.5	10,388	25
Jemez (Pajarito)	NM	continental	8,500–10,441	10	8.2	9,514	5
Klamath & Siskiyou (Mt Shasta)	CA+OR	maritime	5,500–7,800	8	7.7	5,607	4
Madison & Gallatin (Big Sky)	MT	continental	6,800–11,166	10	9.2	8,173	10
Markagunt (Brian Head)	UT	continental	9,600–10,970	13	9.9	9,123	3
Missoula ranges (Montana Snowbowl)	MT	transitional	5,000–7,600	7	6.2	6,013	5
Mt Hood	OR	maritime	4,500–8,540	11	9.4	4,268	2
N Sangre de Cristo (Taos)	NM+CO	continental	9,200–12,481	13	11.0	10,169	12
N Sierra / Tahoe (Palisades/Heavenly/Rose)	CA+NV	maritime	6,200–10,067	27	24.9	7,573	26
North Cascades (Mt Baker)	WA	maritime	3,500–5,089	10	8.6	4,160	5
Park Range (Steamboat)	CO	continental	6,900–10,568	19	16.5	9,355	17
Ruby Mtns & NE Nevada	NV	continental	6,500–10,000	11	10.0	7,866	10
S San Juans (Wolf Creek)	CO	continental	10,300–11,904	16	12.4	10,970	12
S Sangre de Cristo (Santa Fe)	NM	continental	10,350–12,075	8	5.9	10,371	3
S Sierra (Mammoth/June)	CA	maritime	7,953–11,053	7	6.0	8,909	6
S Washington Cascades (Crystal/White Pass)	WA	maritime	4,400–7,012	24	18.8	4,683	12
San Francisco Peaks (AZ Snowbowl)	AZ	continental	9,200–11,500	8	4.8	8,422	1
San Juans (Telluride/Purgatory)	CO	continental	8,725–13,150	23	21.6	10,381	23
Sawtooth & Smoky (Sun Valley)	ID	intermountain	5,750–9,150	21	18.6	7,514	18
Selkirks (Schweitzer)	ID	transitional	4,000–6,400	9	7.2	5,319	8
Snowy Range & Sierra Madre	WY	continental	8,798–9,663	22	18.7	9,475	7
Spring Mtns (Lee Canyon)	NV	continental	8,510–11,290	4	4.0	8,723	3
Tetons (Jackson/Targhee)	WY+ID	intermountain	6,300–10,450	11	9.3	7,874	11
Tushar Mtns (Eagle Point)	UT	continental	9,000–10,600	11	9.4	9,359	7
Uinta Mtns	UT	continental	8,000–11,000	30	27.9	9,345	26
W Central Idaho (Brundage/Tamarack)	ID	intermountain	5,840–7,640	11	9.9	5,937	6
Wasatch (Alta/Snowbird/Park City)	UT	intermountain	6,800–11,000	30	25.2	8,173	28
White Mtns AZ (Sunrise Park)	AZ	continental	9,200–11,100	9	7.9	8,673	2
Whitefish & Flathead Range	MT	transitional	4,464–6,817	12	10.9	5,443	10
Wind River Range	WY	continental	7,000–10,000	27	23.7	8,956	24

LIMITS

What this does not establish

- An r^2 of 15–35 % is real and it is not a forecast. ENSO shifts the odds for a season. Most of the year-to-year variance remains weather.
- Snow depth records begin around 1993. New-snow ratios are calibrated on that period and applied backward, which assumes density has been stable. Depth-based composites rest on roughly 25 winters

against 45 to 60 for the water-based ones.

- **A SNOTEL gauge is not a ski area.** Even with elevation weighting these are nearby mountain stations, and they say nothing about aspect, grooming or snowmaking.
- **Overlapping regions are not fully independent** where their radii intersect, so the sign tests read slightly optimistically.
- **Six inches is a choice.** Regions with few powder days in absolute terms show large percentage swings on small differences.
- **The strength null is bounded by power.** Within-phase tests had about sixteen winters and could only have detected a correlation above roughly 0.65. The claim is that across 2619 chances nothing rose above noise, and the better-powered nested test also found nothing.

Sources – NOAA Climate Prediction Center, Oceanic Niño Index. USDA NRCS Air & Water Database, SNOTEL daily snow water equivalent, snow depth, precipitation and temperature. NOAA NCEI nClimDiv. Method after J. Brown and J. Harper (2026), *Historical evolution of snowpack capacity to buffer rain-on-snow runoff in a large Columbia River headwaters basin*, Hydrol. Earth Syst. Sci. 30, 5735–5748. False-discovery control after Benjamini & Hochberg; field significance after Wilks (2016).

Analysis code and every table in this report are reproducible from github.com/yetiskier/enso-snowpack.